Challenge 1: Titanic - Machine Learning from Disaster



- **Data Dictionary (Titanic)**
- Typical columns in train.csv:
- **Survived** *(target)*: 0 = No, 1 = Yes
- **Pclass**: Ticket class (1, 2, 3)
- **Name**: Full name; includes honorific **Title** (e.g., Mr, Mrs, Miss)
- **Sex**: male/female
- **Age**: Age in years (often missing)
- **SibSp**: # of siblings/spouses aboard
- **Parch**: # of parents/children aboard
- **Ticket**: Ticket ID (can be shared by groups)
- **Fare**: Passenger fare
- **Cabin**: Cabin number (sparse; first letter is the **Deck**)



- In this competition, you'll gain access to two similar datasets that include passenger information like name, age, gender, socio-economic class, etc. One dataset is titled train.csv and the other is titled test.csv.
- Train.csv will contain the details of a subset of the passengers on board (891 to be exact) and importantly, will reveal whether they survived or not, also known as the "ground truth".
- The test.csv dataset contains similar information but does not disclose the "ground truth" for each passenger. It's your job to predict these outcomes.
- Using the patterns you find in the train.csv data, predict whether the other 418 passengers on board (found in test.csv) survived.

Goal

It is your job to predict if a passenger survived the sinking of the Titanic or not. For each in the test set, you must predict a 0 or 1 value for the variable.

Metric

Your score is the percentage of passengers you correctly predict. This is known as accuracy.



Data preview

Numbers of passengers in the train data: 891
Numbers of passengers under the age of 18: 113
Numbers of passengers survived: 342

Survival distribution:
Survived
0 549
1 342
Name: count, dtype: int64
Survival rate: 38.38%

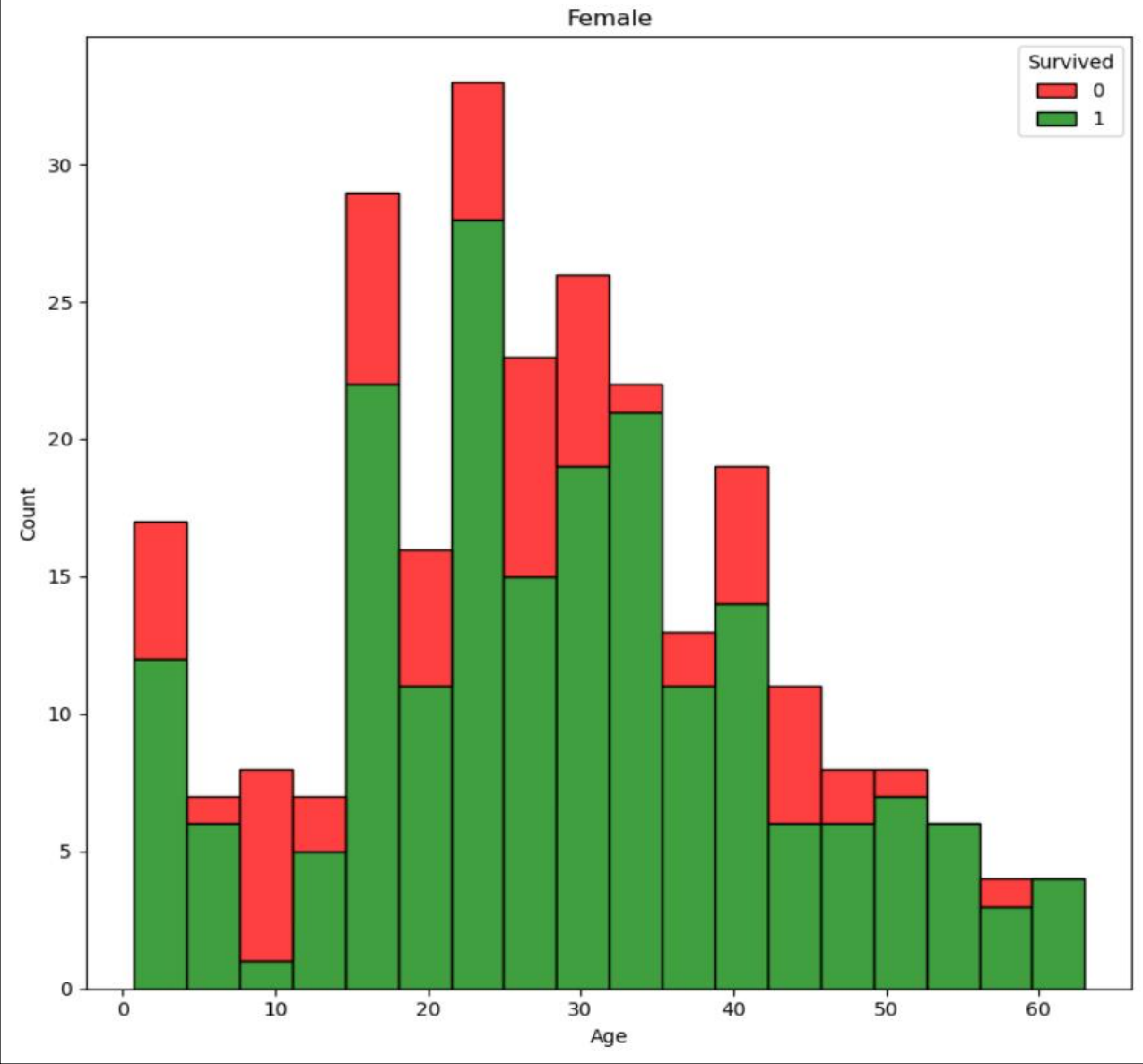
- % of women who survived: 0.7420382165605095

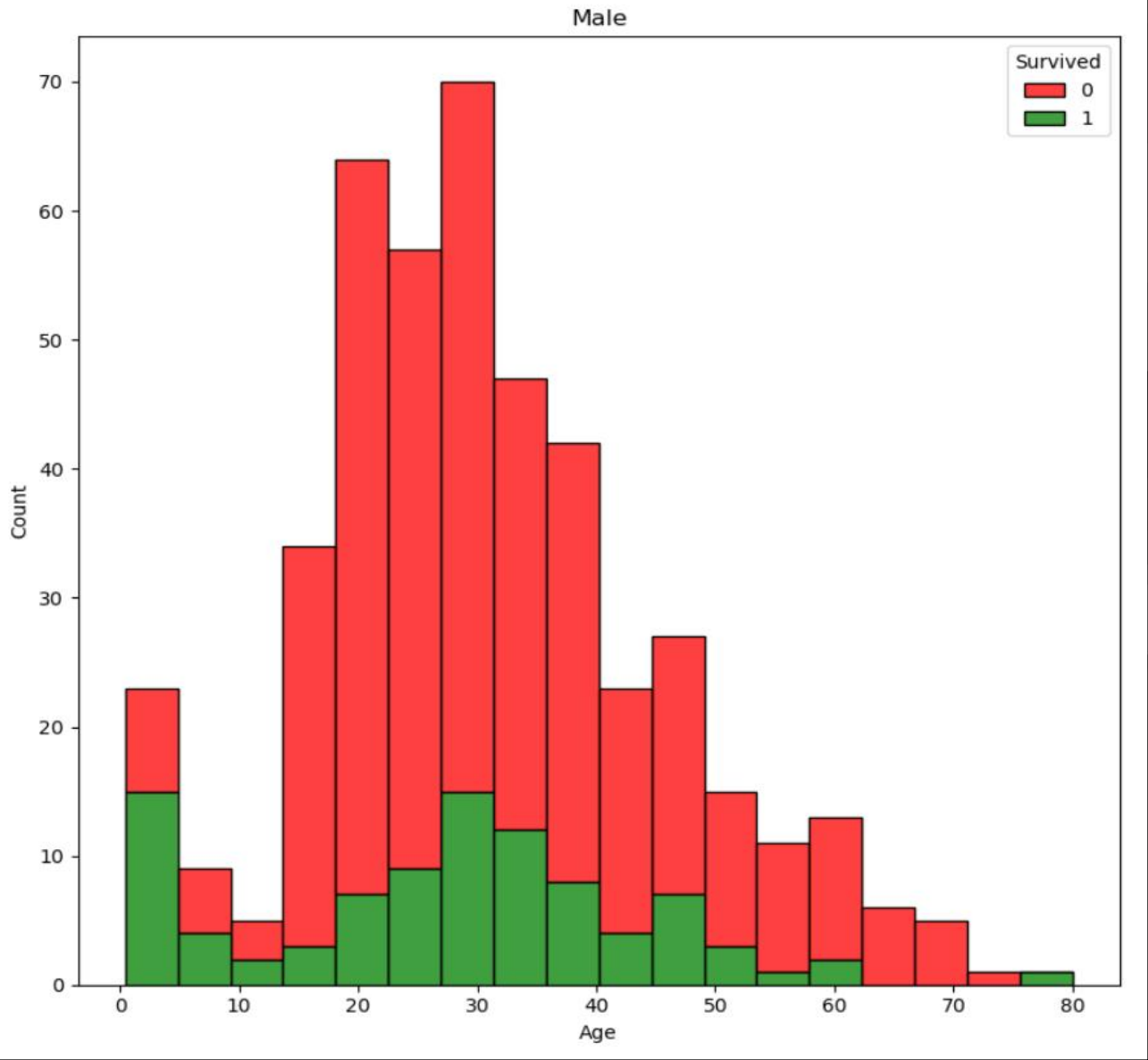
Women and children are the top priority for lifeboat access as the men have given up their places for them, so their chances of survival are higher than those of the men.

- % of men who survived: 0.18890814558058924

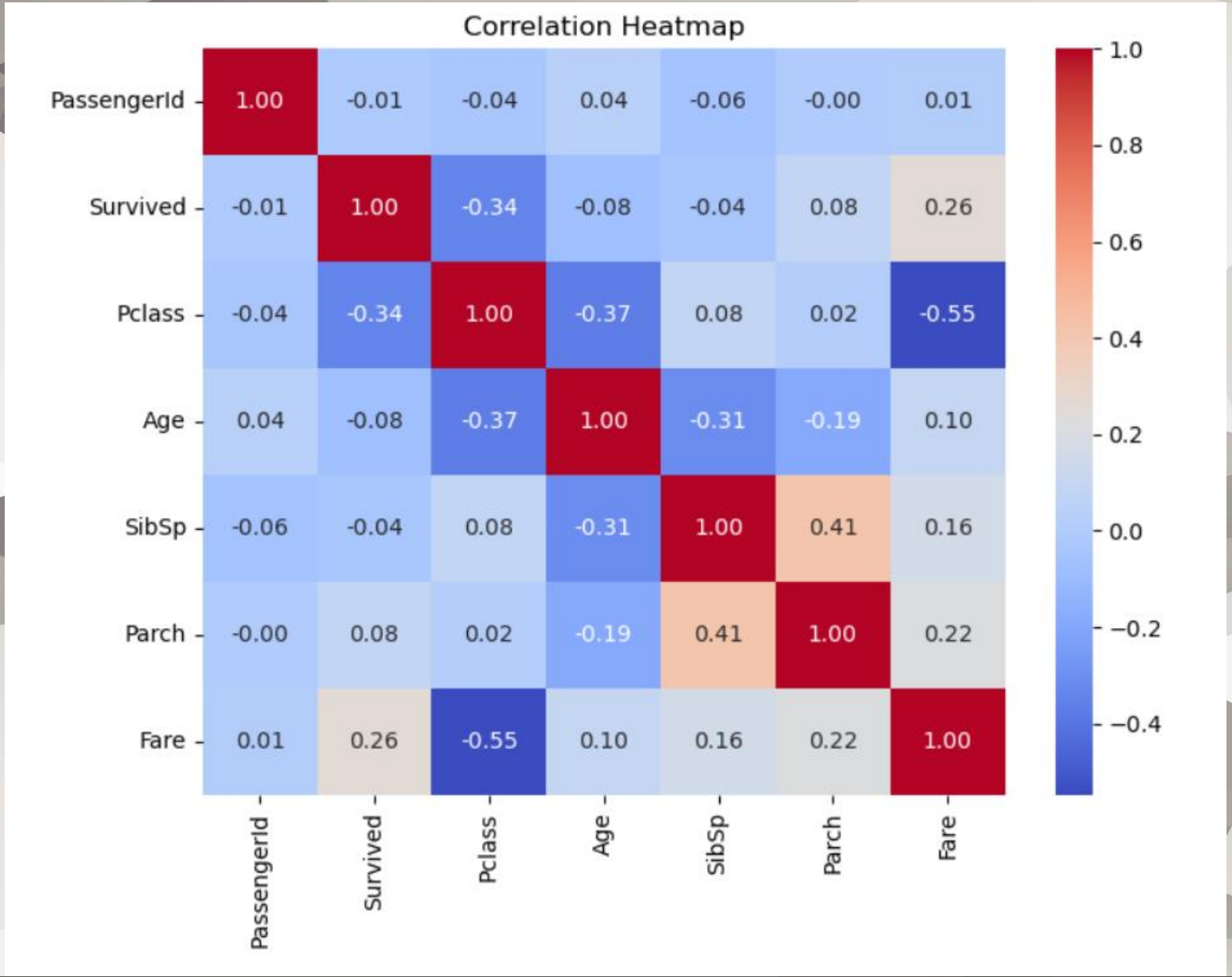
From this you can see that almost 75% of the women on board survived, whereas only 19% of

Visualizing

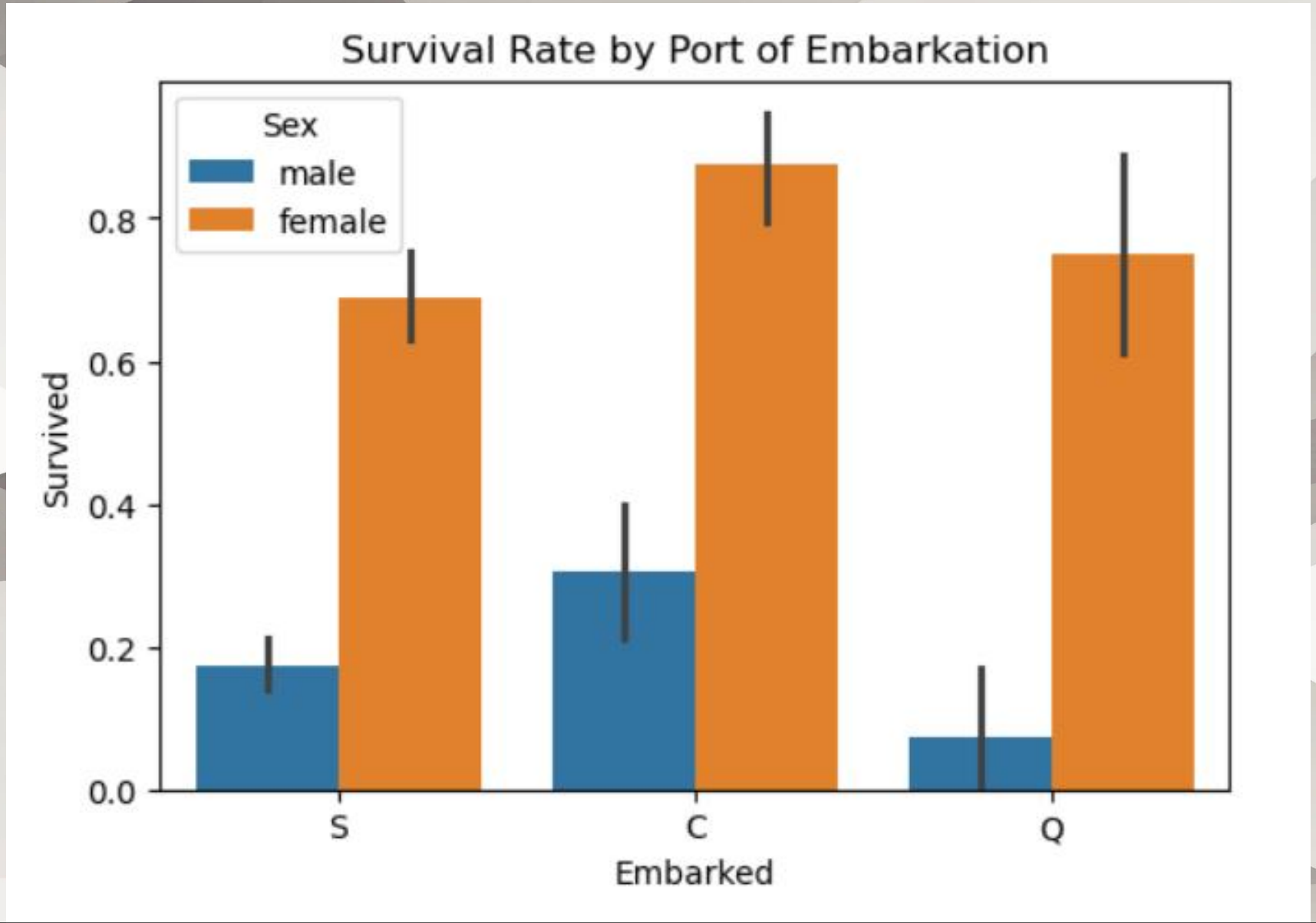




Visualizing



Visualizing



Survival Distribution by Age, Gender, and Passenger Class

The scatter plot below explores how **age**, **gender**, and **passenger class (Pclass)** relate to survival outcomes on the Titanic dataset.

Each point represents a passenger:

- **Red dots (0)** → Did not survive
- **Green dots (1)** → Survived

The plot is separated by:

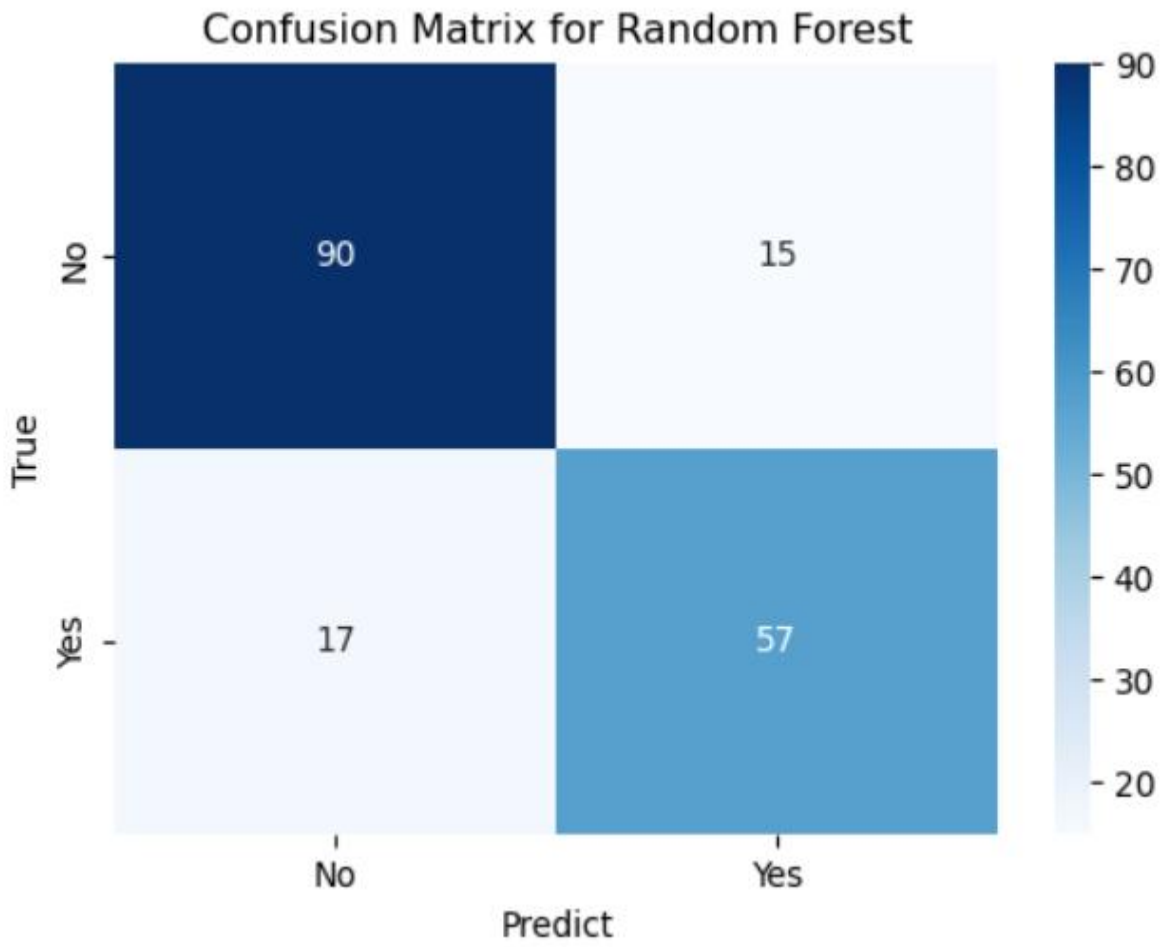
- **Columns** → Passenger Class (Pclass = 3, 2, 1)
- **Rows** → Gender (Sex = male / female)

Observations

- **Females in higher classes (Pclass 1 & 2)** show a much higher survival rate (more green points).
- **Males**, especially in **Pclass 3**, had the lowest survival rate (mostly red points).
- **Younger passengers** have a slightly better chance of survival across all classes — possibly due to evacuation priority for women and children.
- **Older passengers** were less likely to survive, especially men in lower classes.

Insight

This visualization clearly supports the "women and

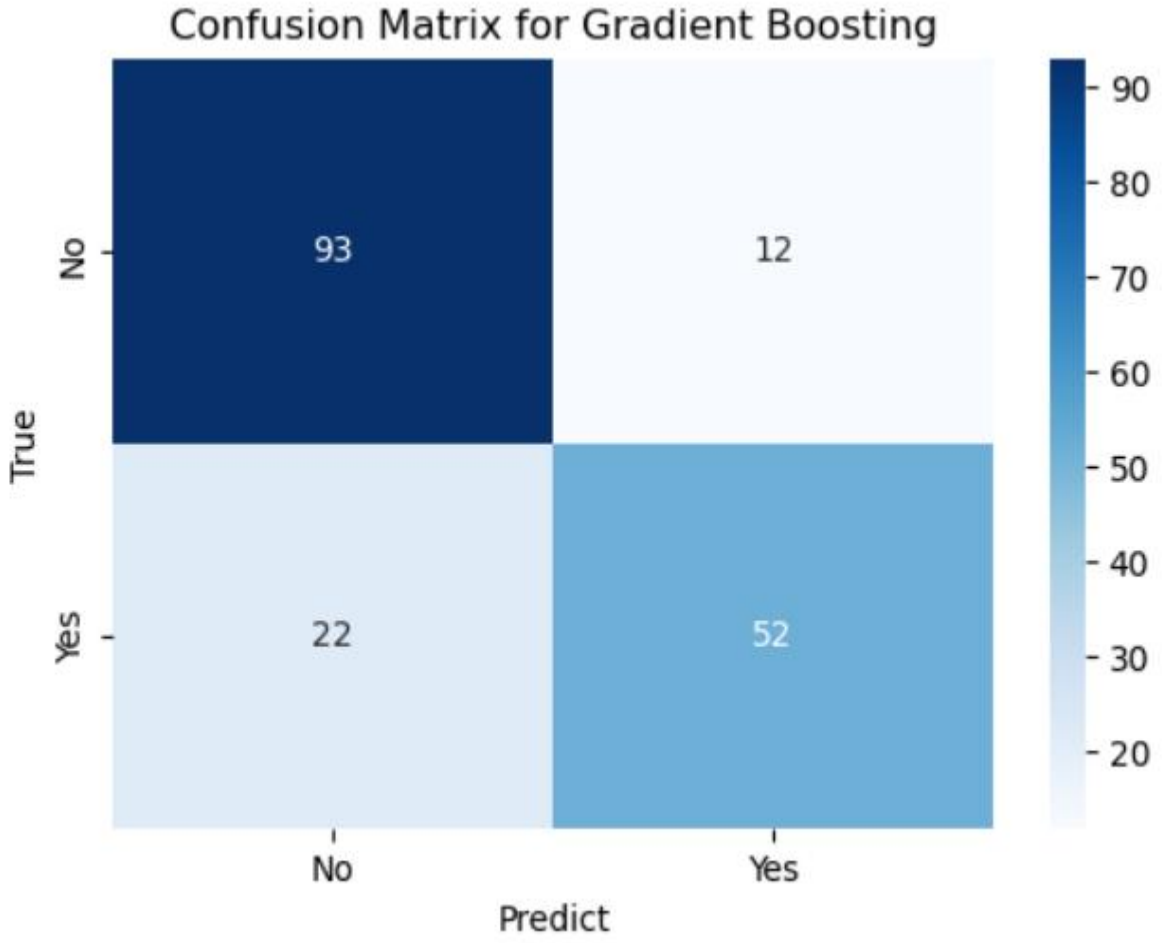


Machine Learning models

```
--- Training Random Forest ---
Accuracy: 0.8212
Classification Report:
              precision    recall  f1-score   support

           0       0.84      0.86      0.85       105
           1       0.79      0.77      0.78        74

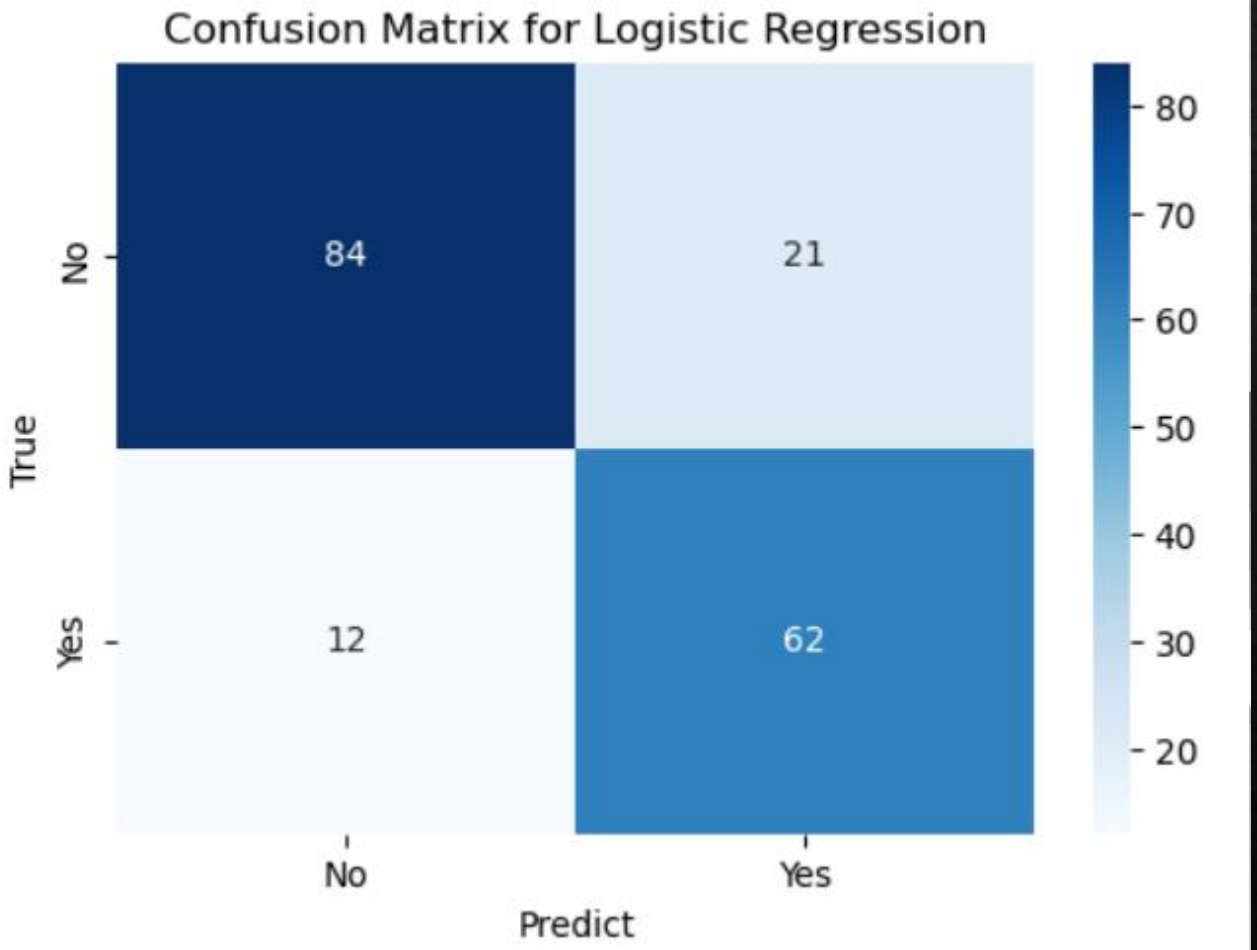
    accuracy                           0.82       179
   macro avg       0.82      0.81      0.81       179
weighted avg       0.82      0.82      0.82       179
```



Machine Learning models

--- Training Gradient Boosting ---
Accuracy: 0.8101
Classification Report:

	precision	recall	f1-score	support
0	0.81	0.89	0.85	105
1	0.81	0.70	0.75	74
accuracy			0.81	179
macro avg	0.81	0.79	0.80	179
weighted avg	0.81	0.81	0.81	

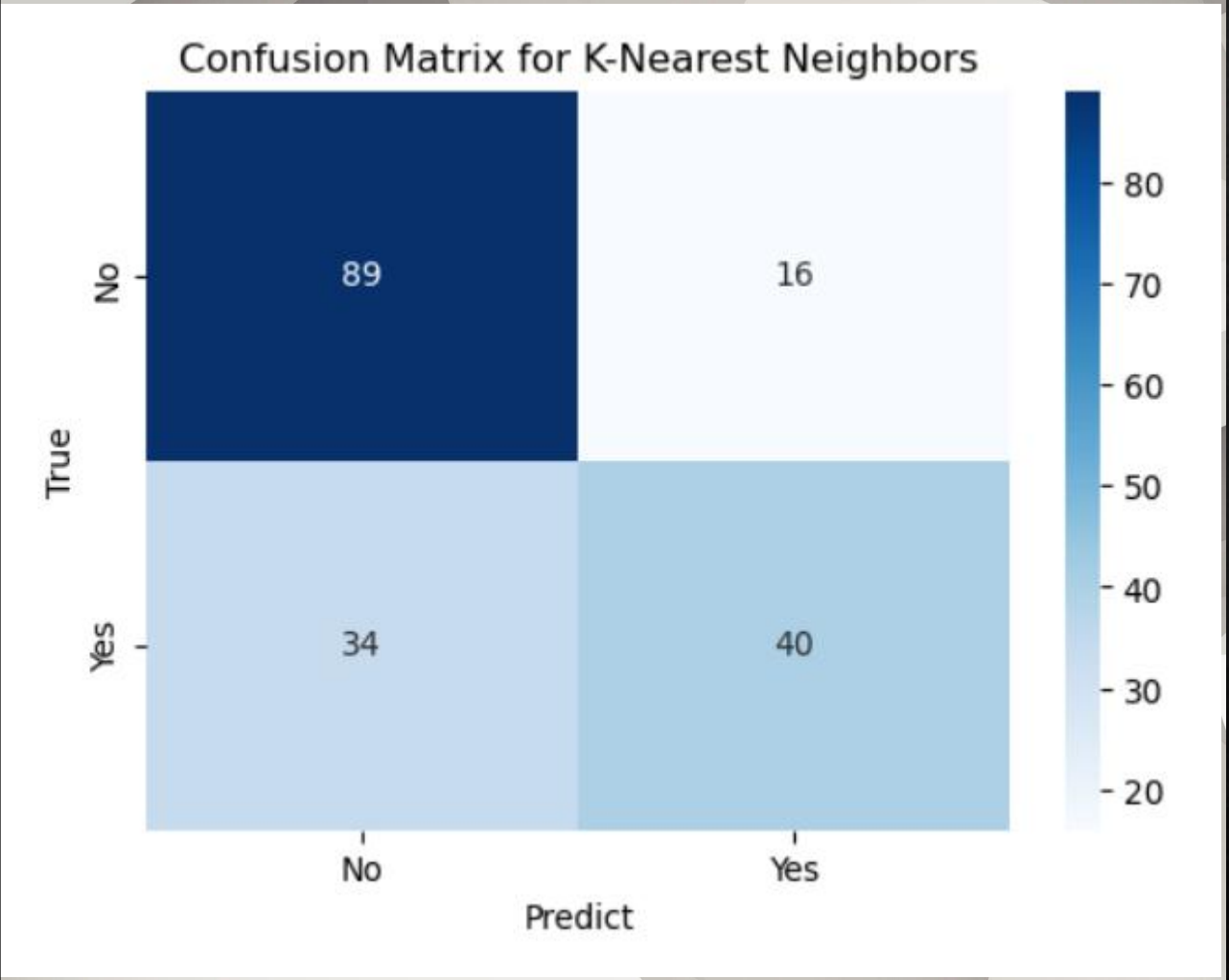


Machine Learning models

```
--- Training Logistic Regression ---
Accuracy: 0.8156
Classification Report:
              precision    recall  f1-score   support

           0       0.88      0.80      0.84       105
           1       0.75      0.84      0.79        74

    accuracy                           0.82       179
   macro avg       0.81      0.82      0.81       179
weighted avg       0.82      0.82      0.82       179
```

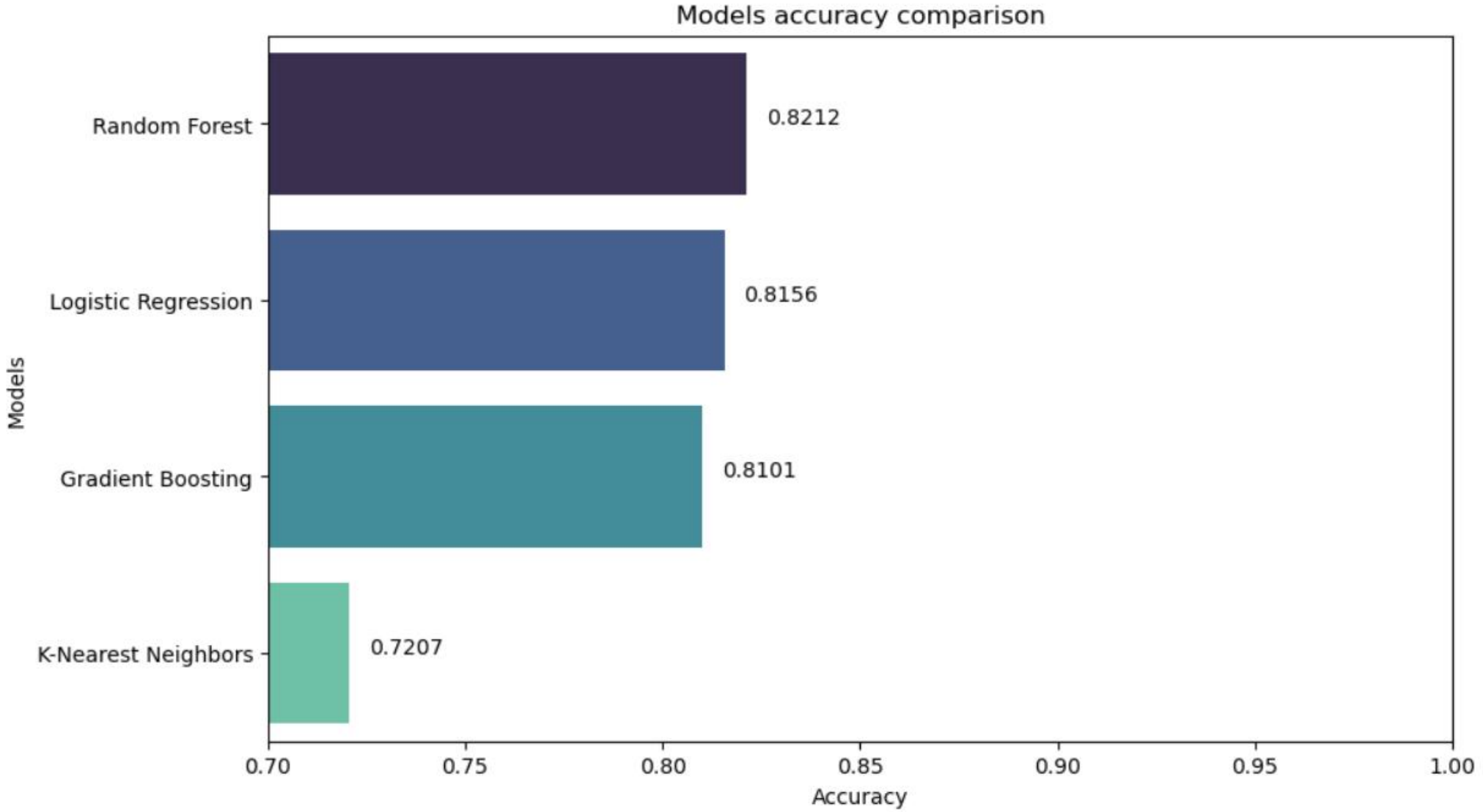


Machine Learning models

--- Training K-Nearest Neighbors ---
Accuracy: 0.7207
Classification Report:

	precision	recall	f1-score	support
0	0.72	0.85	0.78	105
1	0.71	0.54	0.62	74
accuracy			0.72	179
macro avg	0.72	0.69	0.70	179
weighted avg	0.72	0.72	0.71	

Machine Learning models



Analysis of Classification Models on the Titanic Dataset

After training the four models: **Random Forest**, **Gradient Boosting**, **Logistic Regression**, and **K-Nearest Neighbors (KNN)**, we can make the following observations:

1. **Accuracy**
- **Random Forest** usually achieves the highest accuracy due to its ensemble of decision trees and ability to capture non-linear relationships.
- **Gradient Boosting** also performs well, especially when the data has complex feature interactions.
- **Logistic Regression** is a simple linear model, effective when the data is approximately linear, but may have lower accuracy compared to tree-based models.
- **K-Nearest Neighbors (KNN)** depends on the choice of k and feature scaling; it may perform worse if the dataset has many features or uneven distributions.

2. Confusion Matrix

- Shows the number of **correct predictions** (True Positives and True Negatives) and **misclassifications** (False Positives and False Negatives) for each model.
- Random Forest and Gradient Boosting usually have **fewer False Negatives**, meaning they are less likely to miss predicting survivors.
- KNN and Logistic Regression may make more mistakes in certain classes.

3. Classification Report

- Includes **Precision, Recall, and F1-score** for each class:
  - **Precision**: proportion of correct positive predictions among all predicted positives.
  - **Recall**: proportion of correct positive predictions among all actual positives.
  - **F1-score**: harmonic mean of precision and recall.
- Random Forest and Gradient Boosting often have higher F1-scores, indicating a good balance between precision and recall.



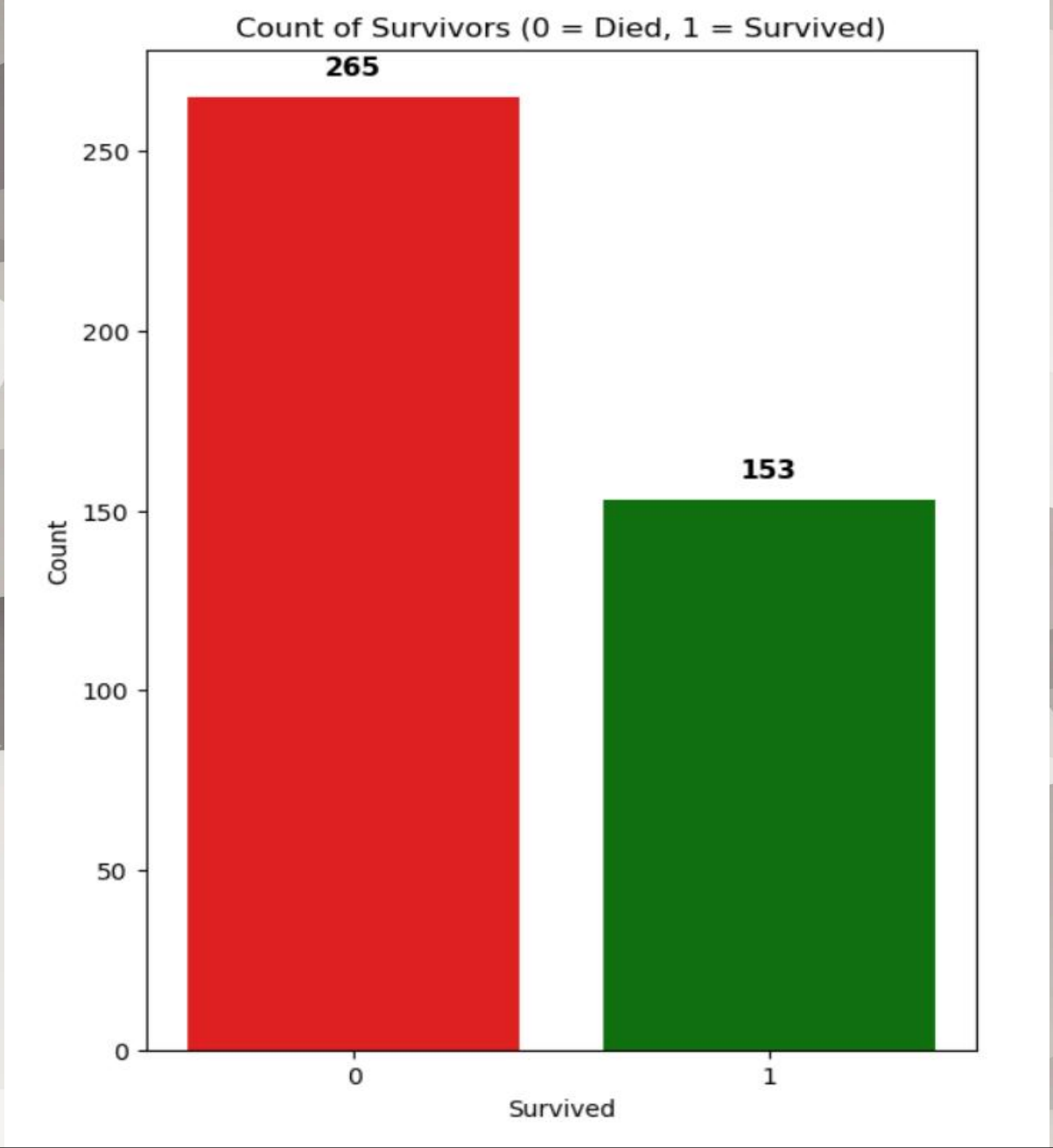
4. Accuracy Comparison Chart

- The bar chart clearly shows the **best-performing model**, typically Random Forest (or Gradient Boosting depending on data).
- KNN and Logistic Regression are slightly lower in accuracy but are still reasonable choices for simpler or faster models.

5. Conclusion

- **Random Forest** is the overall best model on the Titanic dataset due to its ability to handle complex, non-linear data.
- **Gradient Boosting** is a strong alternative, especially when hyperparameters are tuned.
- **Logistic Regression** and **KNN** are suitable for simpler models that are easy to interpret.
- Confusion Matrix and Classification Report provide more detailed insights than accuracy alone.

Machine Learning



THANK YOU